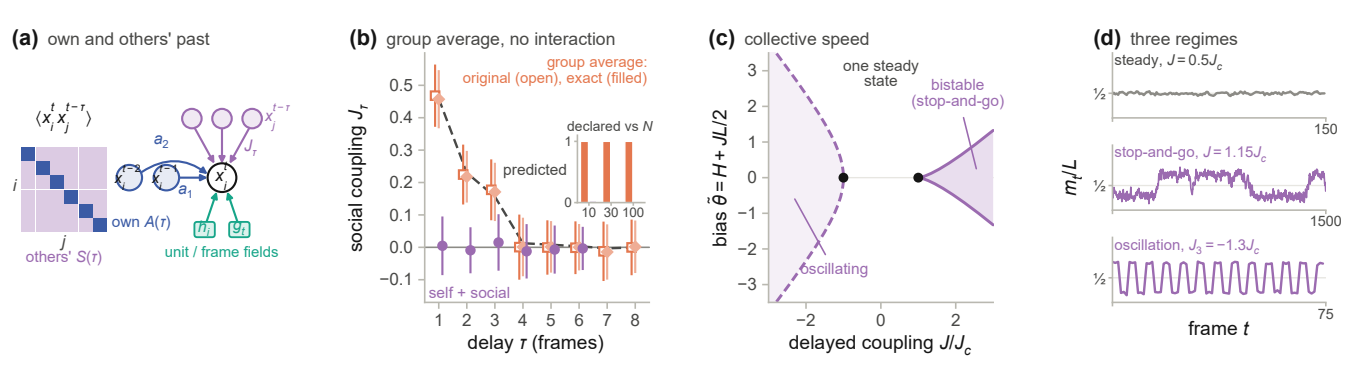

(a) own and others' past
$\langle x_i^t x_j^{t-\tau} \rangle$
i
j
own $A(\tau)$
others' $S(\tau)$
a_2
a_1
J_τ
unit / frame fields
(b) group average, no interaction
group average:
original (open), exact (filled)
predicted
declared vs N
self + social
social coupling J_τ
delay τ (frames)
(c) collective speed
one steady state
bistable (stop-and-go)
oscillating
bias $\tilde{\theta} = H + JL/2$
delayed coupling J/J_c
(d) three regimes
steady, $J = 0.5J_c$
stop-and-go, $J = 1.15J_c$
oscillation, $J_3 = -1.3J_c$
m_t/L
frame t